

Electricity

A Level Physics

Electric current

An **electric current** 电流 is a flow of **charge carriers** 载流子. In a metal the carriers are negative conduction **electrons** 电子; in an **electrolyte** 电解质 they are positive and negative **ions** 离子; in a **semiconductor** 半导体 they may be electrons or "holes" 空穴. The **conventional current** 常规电流 direction is the way **positive** charge would flow —opposite to the real flow of electrons in a wire.

Charge

Charge is **quantised** 量子化: the smallest free unit of charge is the **elementary charge** 基本电荷

$$e = 1.60 \times 10^{-19} \text{ C}.$$

Every free charge in this syllabus is a whole-number multiple of e . The unit of charge is the **coulomb** 库仑, C.

Current as the rate of flow of charge

If charge Q passes a point in time t , the current is

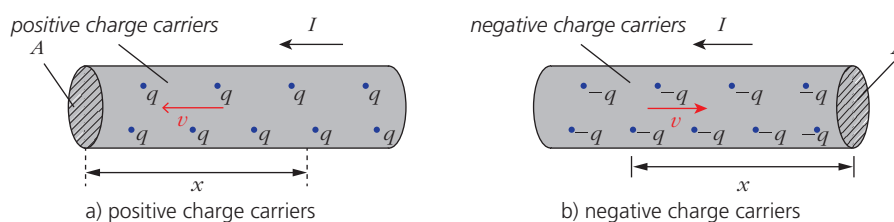
$$I = \frac{Q}{t}, \quad Q = It.$$

Unit of current: ampere, A ($= \text{C s}^{-1}$). For a changing current, the charge that has flowed in a time is the **area under** an I - t graph.

Drift velocity equation

For a uniform conductor of cross-section area A , with n charge carriers per unit volume (the **number density** 数密度), each carrying charge q , moving with average **drift velocity** 漂移速度 v :

$$I = Anvq.$$



Charge carriers drifting inside a conductor

Use this to compare currents:

- a thinner wire (smaller A) at the same I needs a faster drift v .

- a semiconductor has far fewer free carriers than a metal (smaller n), so for the same I the drift velocity is much larger.
- in **series** 串联 components, I is the same everywhere, so if A stays the same but the material changes, nv changes the other way.

Potential difference

The **potential difference** 电势差 (p.d.) across a component is the **energy** 能量 transferred per unit charge as that charge passes through it:

$$V = \frac{W}{Q}.$$

Unit: **volt** 伏特, V (= J C⁻¹).

If 1 J of electrical energy changes into other forms (thermal, light, kinetic, ...) when 1 C of charge passes through a component, the p.d. across it is 1 V.

The **electromotive force** 电动势 (e.m.f.) of a source is the energy given **per unit charge by the source**. The formula is the same as for p.d.; the difference is direction: e.m.f. is energy given **to** the charge by the source; p.d. is energy given **up** by the charge to the component.

Electrical power

Combining $V = W/Q$ and $I = Q/t$:

$$P = \frac{W}{t} = VI.$$

Using Ohm's law $V = IR$:

$$P = VI = I^2R = \frac{V^2}{R}.$$

Pick the form with the quantities you know. Examples:

- two heaters of equal resistance —the one with the larger current gives more **power** 功率 ($P = I^2R$).
- two resistors in **parallel** 并联 across the same **voltage** 电压—the one with smaller R gives more power ($P = V^2/R$).
- a kettle marked "2.4 kW, 240 V" draws $I = P/V = 10$ A and has resistance $R = V^2/P = 24 \Omega$.

Energy transferred in time t is $E = Pt$.

Resistance and Ohm's law

The **resistance** 电阻 R of a component is

$$R = \frac{V}{I}.$$

Unit: **ohm** 欧姆, Ω ($= \text{V A}^{-1}$). Resistance depends on the conditions (such as temperature) when it is measured.

Ohm's law

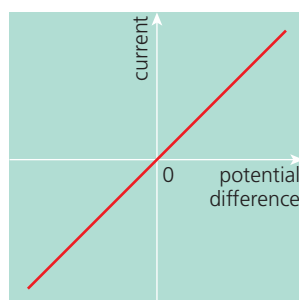
A conductor obeys **Ohm's law** 欧姆定律 when the current through it is **proportional to the p.d.** across it, as long as the conditions (especially temperature) stay constant. For such a conductor R is constant and the I - V graph is a straight line through the origin.

Ohm's law is an experimental result, not a definition. The definition $R = V/I$ works for **any** component; only ohmic ones have constant R .

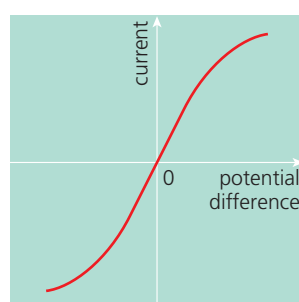
I - V characteristics

You should be able to sketch these:

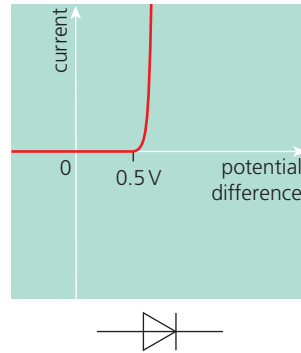
- **metal wire at constant temperature** —a straight line through the origin (constant R). Reversing the p.d. drives the current the other way, giving a straight line in both directions.
- **filament lamp** 灯丝灯泡—through the origin, steep at first, then flatter as V (and I) grow. Reason: more current heats the filament, so its resistance rises and the gradient $1/R$ falls.
- **semiconductor diode** 二极管—almost no current for negative V or small positive V . Above a "switch-on" voltage (about 0.7 V for silicon), the current rises sharply.



I - V characteristic of an ohmic conductor (metal wire at constant temperature)



I - V characteristic of a filament lamp



I–V characteristic of a semiconductor diode

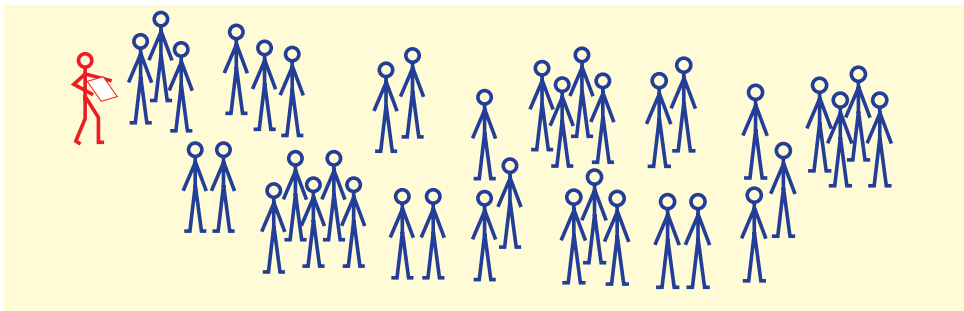
Resistivity

For a uniform conductor of length L and cross-section area A ,

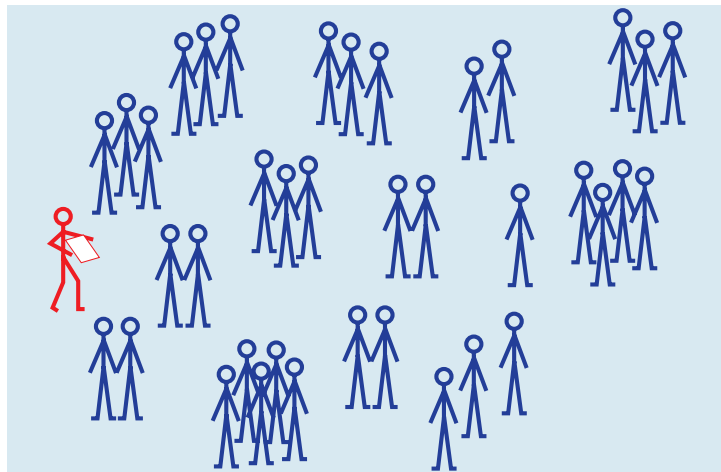
$$R = \frac{\rho L}{A}.$$

ρ is the **resistivity** 电阻率, a property of the material, with unit $\Omega \text{ m}$. Doubling the length doubles R ; doubling the area halves it; halving the diameter quarters the area and so makes R four times bigger.

Typical values: copper at room temperature $\rho \sim 1.7 \times 10^{-8} \Omega \text{ m}$; an **insulator** 绝缘体 $\rho \sim 10^{15} \Omega \text{ m}$ or more.



A longer conductor has more resistance —more collisions along the way

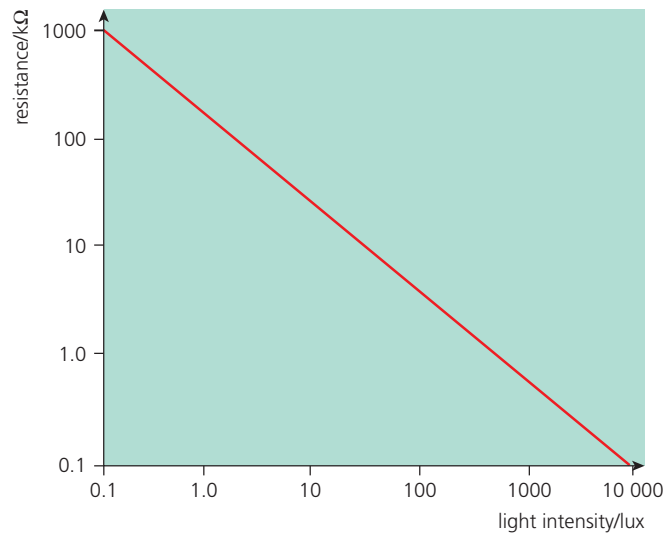


A wider conductor has less resistance —more parallel paths available

The resistivity of a metal **risks with temperature** (more **lattice vibration** 晶格振动 **scatters** 散射 the electrons), which is why the filament lamp's I - V line curves.

Light-dependent resistor (LDR)

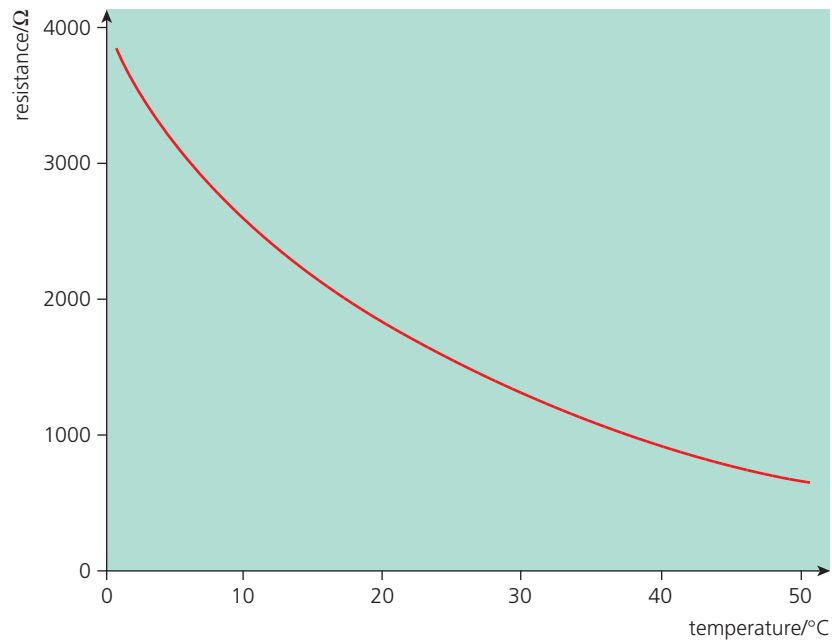
A **light-dependent resistor** 光敏电阻 (LDR) is a semiconductor whose **resistance falls as the light intensity rises**. In bright light R may be a few hundred Ω ; in the dark it can be in the megaohms. LDRs are used in light-sensing circuits (street lamps, camera light meters). Here the **light intensity** 光强 controls the resistance.



Resistance of an LDR decreases as light intensity increases

Thermistor

In this syllabus a **thermistor** 热敏电阻 has a **negative temperature coefficient** 负温度系数: its **resistance falls as its temperature rises**. This is useful for sensing temperature — put it in a **potential divider** 分压器 and the output voltage changes with temperature.



Resistance of a thermistor falls as temperature rises

This is the opposite of a metal: in a semiconductor, more thermal energy frees more charge carriers, and this matters more than the extra scattering.